Exercise 4.5

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
(2)	(2)	(1)	(2)	(4)	(2)	(1)	(4)	(2)	(2)	(3)	(4)	(2)	(3)	(1)
16	17	18	19	20	21	22	23	24	25					
(4)	(2)	(3)	(4)	(3)	(2)	(2)	(2)	(1)	(2)					

Exercise 5.1

(1)(i) $8x^6 - 36x^3 + 54 - \frac{27}{x^3}$ (ii) $2[16x^8 + 216x^4(1-x^2) + 81(1-x^2)^2]$

(2)(i) 108243216 (ii) 96059601 (iii) 4782969

(3) $(1.01)^{10^6} > 10000$ (4) 10 (5) $15, x^6$ term is not possible.

(6) 26235 (7) $-\frac{40}{27}$ (8) $'01'$

(13) $n = 15$ (14) $n = 55$ (15) $n = 7$ or 14

Exercise 5.2

(1)(i) G.P (ii) None of them. (iii) G.P

(iv) None of them (v) None of them (vi) None of them (vii) A.G.P

(2)(i) $2, 2, 4, 4, 6, 6, \ldots$ (ii) $1, 2, 3, 5, 8, 13, \ldots$ (iii) $1, 2, 3, 6, 11, 20, \ldots$

(3)(i) $a_n = \begin{cases} n+1 & \text{if } n \text{ is odd} \\ n & \text{if } n \text{ is even} \end{cases}$ (ii) $a_n = \frac{n}{n+1}, \forall n \in \mathbb{N}$ (iii) $a_n = \frac{2n-1}{2n}, \forall n \in \mathbb{N}$

(iv) $a_n = \begin{cases} 7-n & \text{if } n \text{ is odd} \\ 8+n & \text{if } n \text{ is even} \end{cases}$

(4) $12, 18, 27$ (5) $t_n = \frac{1}{n^2} - \frac{1}{(n+1)^2}$ (8) $5, 45$

Exercise 5.3

(1) $a = \frac{133}{25}, d = -\frac{2}{75}, S_{20} = \frac{304}{3}$ (2) $S_{17} = 527$

(3)(i) $S_n = \frac{8}{81}[10(10^n - 1) - 9n]$ (ii) $S_n = \frac{6}{81}[10(10^n - 1) - 9n]$

(4) $S_n = \frac{4}{9}(4^n - 1) - \frac{n}{3}$ (5) $\frac{3n-2}{3^{n-1}}, \frac{3^n - (3n-2)}{2 \cdot 3^{n-1}} + \frac{3^{n-1} - 1}{4 \cdot 3^{n-3}}$

(6) $n = 15$ (8) 20 months (9) 2480 metres (10) $120, 480, 30(2)^n$

(11) $500\left(\frac{11}{10}\right)^{10} = 1296.87$ (12) 15th day